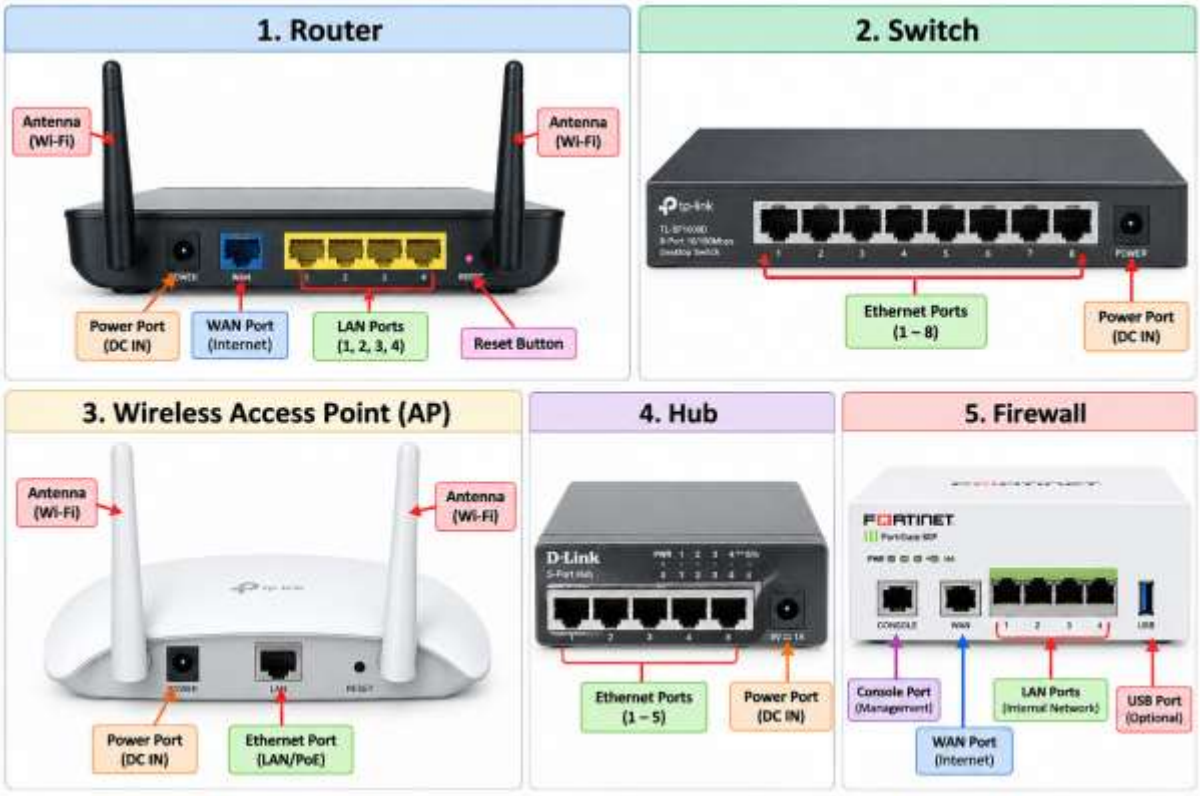
Network Devices Theory

Q1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Networking Device	Main Function	Typical Ports / Interfaces	Use in Home Wi-Fi
Switch	Connects multiple devices within a LAN and forwards data using MAC addresses.	5, 8, 16, or 24 Ethernet (RJ45) ports	Connects computers, printers, and smart TVs.
Router	Connects different networks and directs data using IP addresses.	1 WAN port, 4 LAN ports (typical)	Connects the home network to the internet.
Hub	Connects devices and broadcasts incoming data to all other ports.	4, 8, or 16 Ethernet (RJ45) ports	Legacy device; rarely used in modern home networks.
Firewall	Monitors and filters network traffic to protect against unauthorized access.	WAN and LAN Ethernet interfaces; varies by device	Protects the home network from unwanted traffic.
Wireless Access Point (AP)	Provides wireless connectivity to devices using Wi-Fi.	Usually 1 Ethernet (RJ45) port; sometimes additional ports	Connects smartphones, laptops, and tablets wirelessly.

Q2 Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices..



Device	Ports / Components	Function
Router	WAN, LAN, Power, Antenna	Connects the home network to the internet and provides wired/wireless connectivity.
Switch	Ethernet Ports, Power	Connects multiple wired devices within a LAN.
Wireless Access Point	Ethernet, Power, Antenna	Provides wireless network access to smartphones and laptops.
Hub	Ethernet Ports, Power	Connects devices and repeats incoming data to other ports.
Firewall	WAN, LAN, Console, Power	Filters network traffic and protects the network.

Q3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

Practical: Role of Network Devices in Using Instagram

App Selected: Instagram

1. When I open Instagram on my smartphone, the **Wireless Access Point** connects my phone to the home Wi-Fi network.
2. The **Router** forwards my requests through the internet to Instagram's servers and delivers the responses back to my phone.
3. The **Firewall** helps protect my network by filtering unwanted or unauthorized traffic.
4. These network devices work together to provide secure and smooth access to Instagram for browsing posts, watching reels, and sending messages.

Conclusion: Wireless access points, routers, and firewalls play important roles in providing internet connectivity and network security while using Instagram.

Q4 Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.Think about device capacity, coverage, and traffic handling.

Practical: Troubleshooting Slow College Wi-Fi

Problem: College Wi-Fi is slow and frequently disconnects.
Possible Reason: Too many students may be connected to a single wireless access point, causing network congestion, slow internet speeds, and frequent disconnections.
Suggested Solution: Upgrade the existing access point to a modern Wi-Fi 6 access point with higher device capacity and better traffic management.
Benefits of Upgrading:

1. **Supports more connected devices simultaneously.**
2. **Improves Wi-Fi speed and network performance.**
3. **Reduces congestion and connection drops.**
4. **Provides more efficient wireless communication.**